# Wasti Murad

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## SUMMARY

Recent MSc Data Science graduate specializing in end-to-end machine learning systems, predictive modeling, and big data applications. Experienced in model optimization and interpretable AI solutions, with a focus on automated deployment and experimentation frameworks. Strong background in statistical validation, cloud-based workflows, and data-driven storytelling, translating complex analytical results into actionable insights. A methodical problem-solver who applies the same critical thinking and strategic reasoning to data challenges as to planning a few moves ahead in chess.

## EDUCATION

**Maastricht University**, Limburg, Maastricht, Netherlands — Sep 2023 – Aug 2025
MSc Data Science for Decision Making — GPA 3.7/4.0
*Key courses:* Computer Vision, Deep Learning, Data Mining, Computational Statistics

**Jahangirnagar University**, Savar, Dhaka, Bangladesh — Mar 2017 – Sep 2022
BSc Physics — GPA 3.2/4.0
*Key courses:* Quantum Mechanics, Nuclear Physics, Solid State Physics, Mathematical Physics

## SKILLS

**Programming and Tools:** Python, SQL, Excel, Git, GitHub, Tableau, Power BI
**Machine Learning and AI:** LangChain, RAG, LLMs, PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Transfer Learning, Time Series, Predictive Modeling
**Data Science Libraries:** Pandas, NumPy, SciPy, Statsmodels, OpenCV
**Big Data Processing:** Apache Spark, PySpark, Spark Optimization (Partitioning, Broadcast Joins, Caching, AQE)
**Data Engineering:** Databricks, Delta Lake, Delta Live Tables, Medallion Architecture, Unity Catalog
**Cloud and MLOps:** Azure Data Factory, AWS, SageMaker, MLflow, Hydra, Docker, GitHub Actions, CI/CD
**Experimentation and Analytics:** A/B Testing, Statistical Testing, Power Analysis, Sequential Testing
**Languages:** Bengali (Native), English (C1), Dutch (A1)
**Professional:** Communication, Teamwork, Project Management, Reproducibility, Time Management

## EXPERIENCE

**Master's Thesis – Artificial Intelligence Guided Adaptive Radiotherapy Using Counterfactual Explanations in Latent Space**

*Maastricht University and Maastro (Clinical Data Science)* — *Maastricht, Netherlands* — Feb 2025 – Aug 2025

- Built a 3D Variational Autoencoder for anatomy-preserving counterfactuals, addressing interpretability gaps in radiotherapy AI and improving reconstruction by 4%
- Restructured the latent space to disentangle tumor from organ variations, enabling clinician-validated traversals for medical review
- Established a reproducible experimentation workflow with consistent run tracking, enabling collaborative research and reliable cross-team result comparison

*Technologies: Python, PyTorch, CUDA, MONAI, MLflow*

**Machine Learning Intern – Model Generalization and Data Pipelines**

*Maastricht University Medical Center* — *Maastricht, Netherlands* — Sep 2024 – Feb 2025

- Enhanced preprocessing and class-balancing pipelines to stabilize training on imbalanced medical datasets, improving model reliability and generalization on unseen patient data
- Developed a configuration and experiment-tracking framework, replacing fragmented setups and manual logging, enabling reproducible experiments and smoother team collaboration
- Produced clinician-friendly Explainable AI summaries, bridging technical outputs and clinical interpretations, enabling stakeholder-ready insights

*Technologies: Python, Pandas, AutoML, Hydra, MLflow, Git, GitHub*

## RESEARCH EXPERIENCE

**Personal Wildlife Monitoring** (Group Research Project)

*Maastricht University* — *Maastricht, Netherlands* — Aug 2023 – Jul 2024

- Architected an edge-to-cloud pipeline with motion and audio triggers, automating manual stream screening
- Integrated object detection, Transformer models, and acoustic features into a multimodal system, cutting identification time from hours to minutes
- Deployed the system behind a containerized API, enabling continuous real-time monitoring over post-hoc review

*Technologies: Python, OpenCV, Librosa, Transformers, FastAPI, Docker*

## TECHNICAL PROJECTS

**Code Repository**

### Causal Inference and Uplift Modeling for Marketing Campaign Optimization

- Estimated individual treatment effects from observational data, correcting for selection bias to capture true incremental customer lift
- Identified a 25–30% high-uplift segment concentrating most of the campaign's incremental revenue potential
- Designed a Target / Don't-target / Exclude strategy, shifting allocation from broad targeting to profit-driven spend

*Technologies: Python, Pandas, Scikit-learn, Statsmodels, CausalML, Propensity Scoring, T-Learner, Uplift Random Forest*

### LLM-Powered E-Commerce Analytics Assistant

- Architected a hybrid query routing system across structured and unstructured sources, eliminating hallucinated metrics in answers over sales and review data
- Designed an end-to-end evaluation framework measuring faithfulness and relevance, accelerating iteration and closing reliability gaps in LLM analytics
- Delivered an interactive interface for non-technical users to query sales and reviews via natural language

*Technologies: Python, LangChain, OpenAI, RAG, Pandas, Streamlit, YAML*

### End-to-End Machine Learning Pipeline with Automated Training and Deployment

- Built a CI/CD pipeline automating model training, testing, and Docker image publishing on each commit, replacing manual workflows with reproducible releases
- Implemented unit and integration tests to catch regressions early and ensure production-ready builds
- Versioned model artifacts and container images to enable traceable releases and safe rollbacks

*Technologies: Docker, scikit-learn, GitHub Actions, CI/CD, MLOps, pytest*

### Apple Retail Sales and Warranty Analytics

- Surfaced revenue concentration in multi-year Apple retail data, showing the top 20% of stores drive 40% of revenue to guide store prioritization
- Evaluated product reliability via 180-day warranty cohorts, finding mid-tier products had 35% fewer claims than premium tiers
- Delivered Tableau dashboards to track regional trends and support data-driven store and product decisions

*Technologies: SQL, Window Functions, CTEs, Joins, Tableau*

### Databricks Delta Live Tables: Streaming ELT Medallion Pipeline

- Engineered a streaming ELT pipeline with Bronze/Silver/Gold layers, replacing brittle batch workflows with schema and quality enforcement at every stage
- Designed transformation logic for joins, derived metrics, and type casting to produce analytics-ready datasets
- Versioned the full pipeline in Git and Databricks, enabling reproducible and peer-reviewable data engineering workflows

*Technologies: Databricks, PySpark, Delta Live Tables, Delta Lake, Medallion Architecture, SQL*

## AWARDS

**International University Physics Competition** 2019, 2020

- Bronze Medal (2020): Problem A – Protecting Travelers to Mars
- Silver Medal (2019): Problem A – Ion Thrusters to Saturn